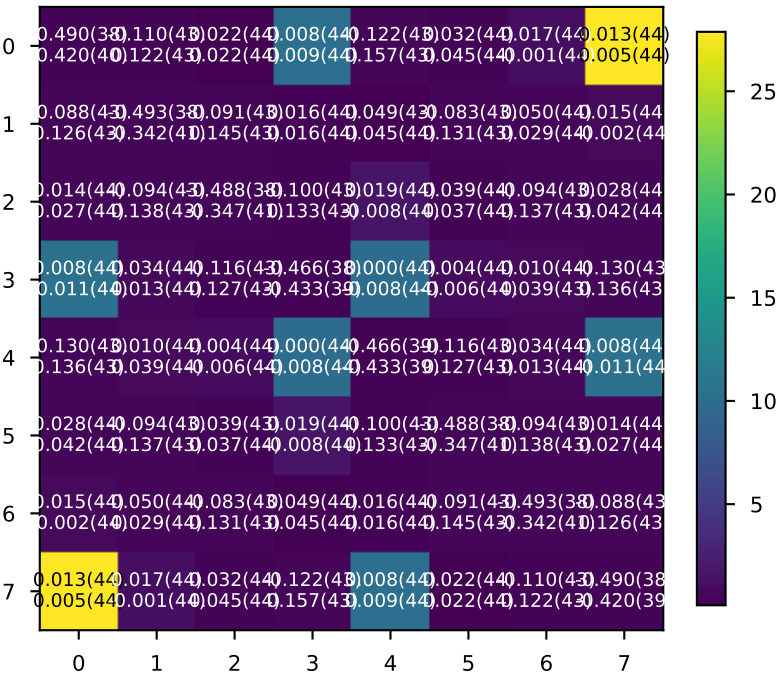
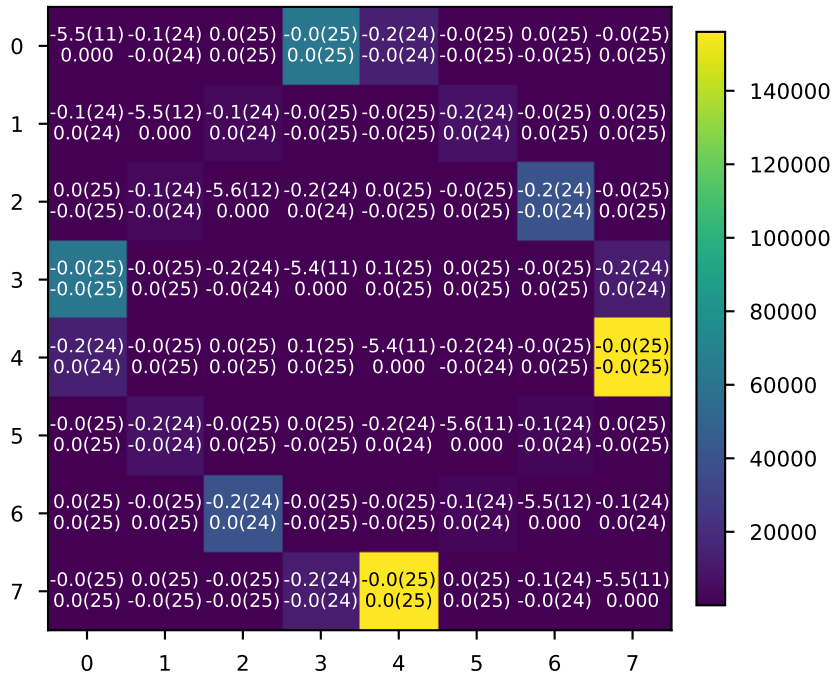
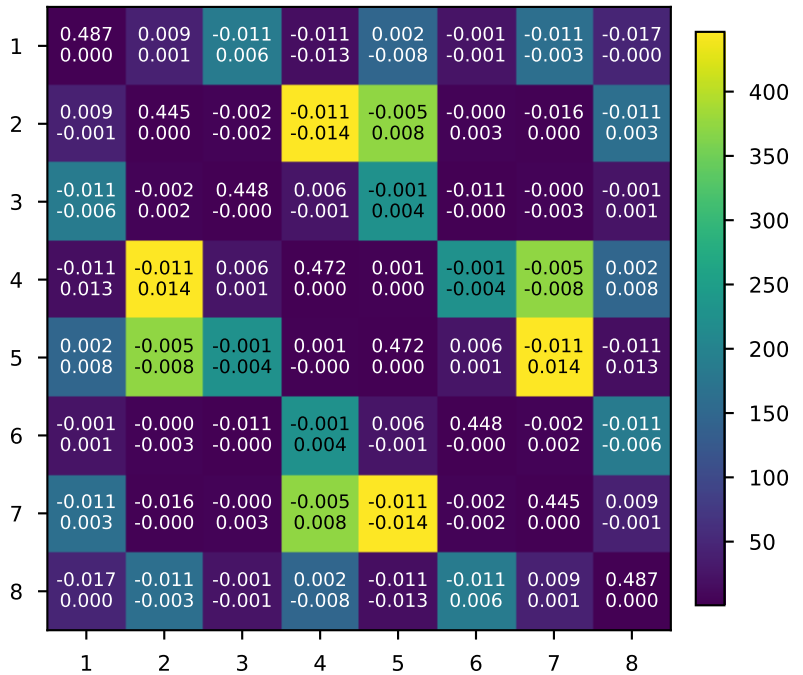
4 qubits - $\langle j|U|k\rangle$



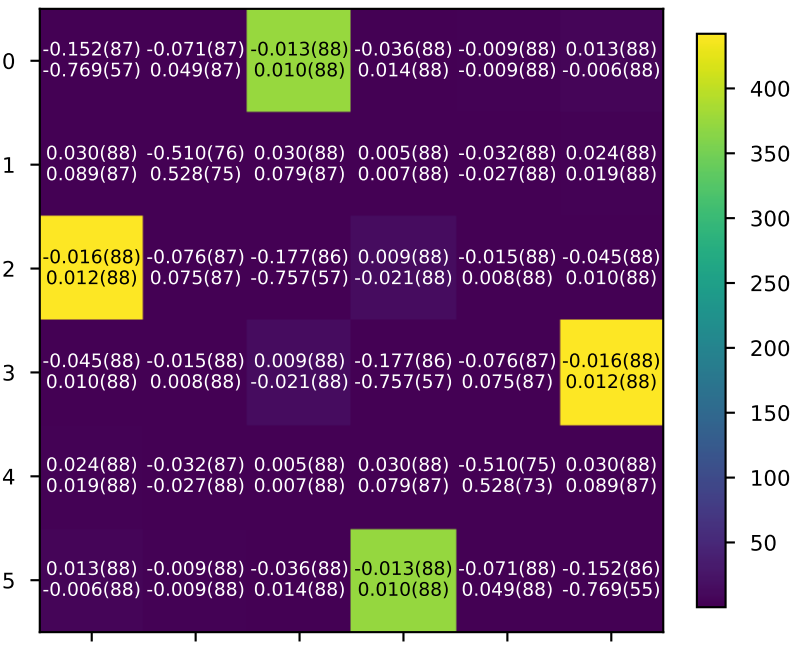
4 qubits - $\langle j|U^\dagger HU|k\rangle$



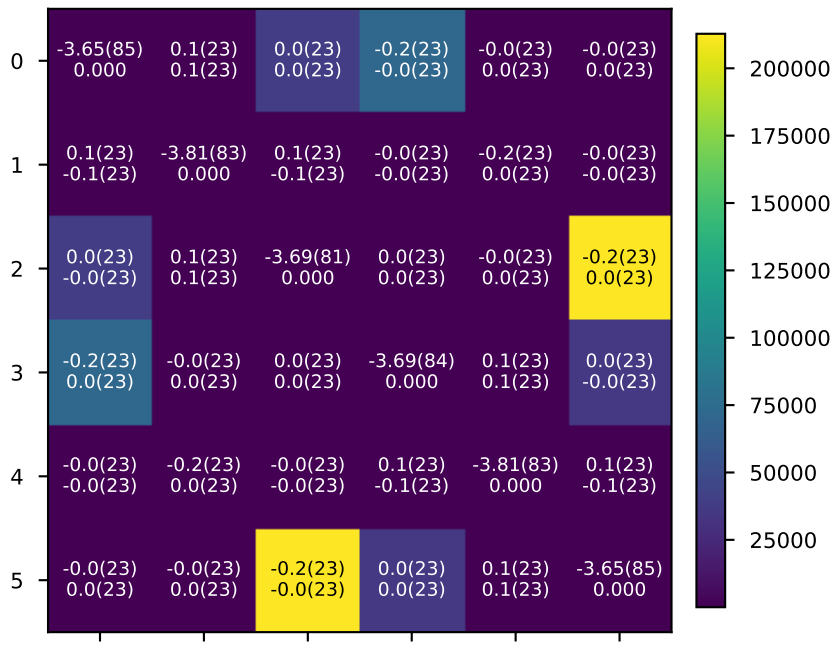
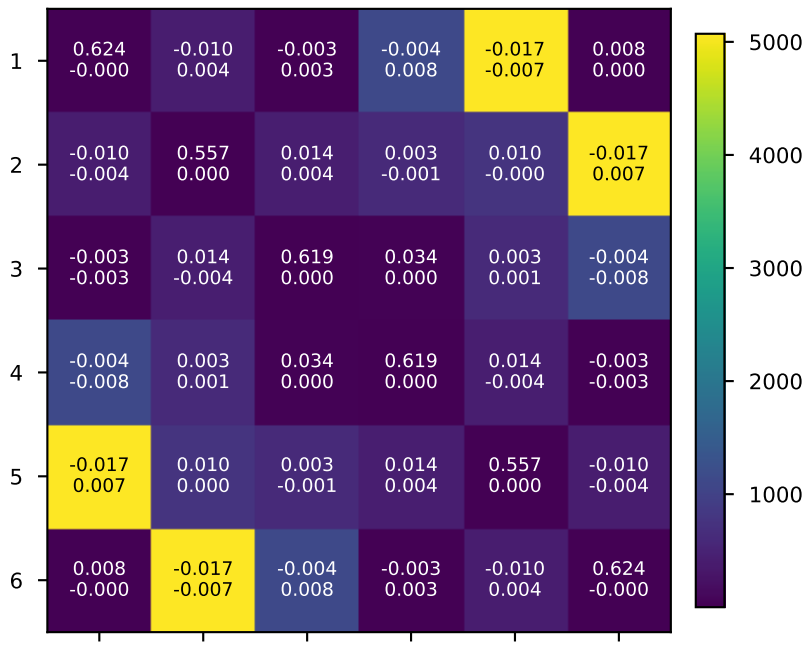
4 qubits - $OO^\dagger$



3 qubits - $\langle j|U|k\rangle$



3 qubits - $\langle j|U^\dagger HU|k\rangle$



3 qubits - $OO^\dagger$